Methods for fitting the model

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Abstract

1 Introduction

20 parameters from the mitotic module were chosen, previously reported to be sensitive to intermitotic times [ref]. Not that all parameters values were taken from (or close to?) models implemented by Goldbeter. We also chose the all 10 parameters from the apoptosis module. We decided to focus on a global sensitivity analysis of those parameters regarding to the IMT distributions and the fractional proliferation model parameters calculated from the experimental data.

2 Sobol sequence

The Sobol sequence generation of parameters allowed us to explore the sensitivity of the model regarding 30 his parameters. The sobol sequence [ref]. The advantage is that we can always increase the finesse of the search afterwards in a simple way by generating the sequence of the sequence unlike a simple grid search.

The Sobol sequence were computed in the log scale of the biochemical parameters. The dimensions were centered using the original parameters values, taken from [ref] and range from -2 to +2 the original location in the log space. 120.000 sets of parameters were generated using the Sobol sequence.

3 Cost function structure

3.1 Pseudo code for the cost function

Algorithm 1: Cost function

Checkpoint: do parameters allow the mitotic cycling?
-Fast fourier transform on the ODE simulation for Cyclin E, Cyclin D and Cdc20 and check the strongest non zero frequency term belongs to $\mathrm{Hz}^{-1}(between XX and YY hours)$
-Do peakfinder algorithm can find peaks of Cdc20 and Cyclin D alternate
Run the cell population simulation given a biochemical model of apoptosis and cell cycle
Calculate the death rate, quiescence rate and dividing rate given the fractional proliferation model and the simulated data
Cost = mmd(simulated IMTs, experimental IMTs) +
$|log\ a_{exp} - log\ a_{sim}| + |log\ d_{exp} - log\ d_{sim}| + |log\ q_{exp} - log\ q_{sim}|$

3.2 Population simulation Pseudocode

The aim is using the biochemical and simulating different cells and keeping tracks of cell events like death or mitosis. Then we can used these data to compare it precisely with the data from the experiments.

Algorithm 2: Cell population simulation

```
Run the ODE biochemical model
if The dividing system is not cycling or the protein concentrations are going to high then
    Cost = inf
else
    Take the possible concentrations for each proteins between two peaks of the cyclewhile do there
    are living cells within the experiment time length
    for each living cells do
        Run the Stochastic simulations for 20 hours
        if there is more than 20 BAXm proteins for more than x minutes then
            The cell dies
        else if One Cdc20 peak is detected then
            The cell divides
        else
            The cell is quiescent
        end if
    end for
  end while
end if
```

3.3 Fractional proliferation model

(taken from methods from fractional proliferation paper)

For the different cells we have both mitotic lifespans and censored lifespans, those are the lifespans we know the start but not the end because they are after the end of the experiment, so they may be after the end of the experiment or cells that switched quiescent.

Likelihood of Quiescence model

We find the best parameters for the quiescence model by minimizing the log of the likelihood of fractional cells and quiescent cells given by

$$log(\prod f(x_i, y_j|\mu, \sigma, \lambda)) = log(\prod_i (1-Q) f(x_i|\mu, \sigma, \lambda) \times \prod_j (Q + (1-Q)F(y_j|\mu, \sigma, \lambda))) \tag{1}$$

With x_i are the observed lifespans, these correspond to cycling cells and f gives the probability, given the Exponentially-Modified Gaussian (EMG) model that this intermitotic time happens. And y_j are the censored lifespans, meaning we only observe their birth time during the experiment, thus they may be quiescent cells (with a probability of q) or cycling cells (with a probability of $(1-q)F(y_j|\mu, \sigma, \lambda)$).

Thus we find the best μ, σ, λ and q.

The methods used for minimizing is SLSQP (Sequential quadratic programming) from minimize in the scipy package.

Death rate (s^{-1})

The death rate is calculated with the following formula:

$$a = \frac{Number\ of\ deaths}{Total\ observed\ time}\ (s^{-1}) \tag{2}$$

(In the original paper this is estimated by fitting the log2(proliferation curve))

Division Rate (s^{-1})
The division rate is calculated with the following formula:

$$2\int_0^{\infty} e^{-d\ \tau} EMG(\tau|\mu,\sigma,\lambda)d\tau = 1 \tag{3}$$

Finding this using brentq algorithm from Scipy (Brent's method to find a zero of the function f) and d as a variable

Quiescence Rate (s^{-1})

$$q = \frac{Q}{1-Q} d\ (s^{-1}) \tag{4}$$

3.4 Maximum mean discrepancy

The Maximum Mean discrepancy is a metrics used to measure the difference between two distributions $\mathbb{P}$ and $\mathbb{Q}$. It takes advantage of the kernel trick. The MMD is defined as follow : (ref)

$$M_k(\mathbb{P},\mathbb{Q}) = \|\mu_{\mathbb{P}} - \mu_{\mathbb{P}}\|^2_{\mathcal{H}} \tag{5}$$
$$= \mathbb{E}_{\mathbb{P}}[k(x,x')] - 2\mathbb{E}_{\mathbb{P},\mathbb{Q}}[k(x,y)] + \mathbb{E}_{\mathbb{Q}}[k(y,y')] \tag{6}$$

However, in our case we have a one dimensional data set (set of intermitotic time) and we use the Radial basis function for the kernel: